

Ledgerix Pro - Enterprise Architecture

v3.4

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Railway Cloud Edition — 18-Agent AI Workforce Version 3.4 — May 24, 2026 — CONFIDENTIAL

Property	Value
Founder & Business Strategist	Scott Hansbury
Headquarters	Scottsdale, Arizona
Version	3.4 — May 24, 2026
Horizons Complete	H1, H2, H3, H4 (all in-scope items)
Active Agents	18 (of 25 total configured)
Scheduled Routines	17 cron routines live on Railway
Infrastructure	Railway cloud (api.ledgerixpro.com)
Beta Clients	None (pre-beta — pipeline ready, no active beta clients pending Pattern B Full Phase 5)

1. System Overview

Ledgerix Pro is an AI-powered, autonomous bookkeeping and accounting firm built on the Paperclip AI agent orchestration platform. Deployed on Railway cloud infrastructure, with GoHighLevel (GHL) as the CRM and client data hub. Designed for multi-tenant security, Human-in-the-Loop oversight, and zero data cross-contamination between client workspaces.

1.1 Deployment Environment

Component	Details
Runtime Environment	Railway cloud — <code>api.ledgerixpro.com</code>
Agent Orchestration	Paperclip AI (<code>ledgerix-pro-core</code>) — Railway container
Database	External PostgreSQL on Railway — <code>turntable.proxy.rlwy.net:32057</code>
DB Backups	Automated hourly, 7-day daily, 4-week weekly, 1-month monthly retention
CRM Platform	GoHighLevel (GHL) — LIVE
Webhook Bridge	<code>api.ledgerixpro.com/api/webhooks/ghl</code> — LIVE (no ngrok)
Accounting Integrations	Xero — LIVE (multi-tenant per-contact OAuth). QBO — built, pending client.
Client Portal	<code>api.ledgerixpro.com/portal/{contactId}</code> or <code>/portal/{slug}</code>
Internal Dashboard	<code>api.ledgerixpro.com/dashboard</code> (secret-protected)
Diagnostic Calculator	<code>api.ledgerixpro.com/diagnostic</code> (public CRO funnel)
Company UUID	<code>f60117de-1131-433c-934f-3fe88bfaa163</code>
GHL Location ID	<code>GhnRONQQVJiCKsdWoQFc</code>
Board API Key	<code>pcp_board_railway_admin_key_2026</code>

1.2 Agent Workforce Summary

Division	Active Agents	Dormant / Deferred
Accounting (CFO)	Senior Bookkeeper, Ledger Specialist, Reconciliation Agent, AP Specialist, AR Specialist, Payroll, Reporter, Tax Liaison, Billing & Invoicing	Client Health Monitor (5+ clients)
Operations (COO)	Sentinel, Onboarding, Quality Control, Audit & Compliance, Knowledge Base Manager	—
Revenue (CRO)	SDR (Laura), Client Success Manager, Referral & Reviews, Reactivation	Sales Outreach (50+ leads)
Executive	CEO, CFO, COO, CRO (configured, strategic governance)	—

1.3 Horizon Completion Status

Horizon	Status	Key Deliverables
H1 — Foundation	Complete	GHL webhook pipeline, Dispatcher, workspace registry, dual-path auth
H2 — Accounting Core	Complete	QBO + Xero OAuth, Sentinel, Ledger Specialist, Reconciliation, Senior Bookkeeper, KB Manager
H3 — Operations	Complete	Internal dashboard, weekly digest, budget guardrails, CRO funnel, nurture sequences, SDR (Laura), Client Health Monitor, Reactivation
H4 — Full Platform	Complete	Client portal (slug-based), AP Specialist, Tax Liaison, Reporter, Billing & Invoicing, Payroll, Quality Control, Audit & Compliance, Referral & Reviews, AR payment date intelligence, Railway migration, multi-tenant OAuth
H4 Deferred	Scale-triggered	Sales Outreach (50+ leads), Scale Pattern (50 clients), KB rules DB table (50 clients)

1.4 Trust Tenet (established 2026-05-24)

No real clients — beta or paying — including Ledgerix Pro’s own books are onboarded until the system is correct, trustworthy, and dialed in for security and safety of client funds. This applies uniformly. There is no “ship something today” pressure on safety-critical work; sessions are checkpoints in continuous architectural work, not deadlines for deliverables. Partial-spec compliance on write endpoints touching financial records is rejected as a category — even when it would unblock other work.

Concretely, this means:

- Write endpoints affecting QBO/Xero books cannot ship until the Phase 4c safety architecture is complete
- Audit log entries cannot have placeholder values (e.g., `previousAccountRef: null`) that hide unknown context
- When the test of an endpoint says “X must be captured,” X must actually be captured — not nulled out

Operating Principles (canonical in repo root `CLAUDE.md`)

1. **Verify before assuming.** Grep for callers before claiming a function is unused. View existing imports before claiming code compiles. Read the authoritative document before quoting values from session memory. Skipping these checks has caused real production-shaped errors.
2. **Session-end documentation discipline.** Every session ends with EA + Brief + relevant trackers + WIP docs all reflecting what was committed. No exceptions. Documents of truth must actually be true at session close.
3. **Locked decisions stay locked.** Once an architectural decision is committed (in ADR, WIP doc Decisions Made, or explicit acknowledgment), it does not get reopened mid-session for refinement. Implementation details discovered during execution are handled during execution, not by reopening the decision.
4. **WIP docs are TRUTH for active work.** If a WIP doc and another doc disagree about an in-flight piece of work, the WIP doc is correct.

Multi-Session Continuity: `docs/wip/` **Convention**

Substantial architectural work that spans multiple sessions has a Work-In-Progress document at `docs/wip/<feature-or-phase>.md`. Required structure: Status, Started, Last-updated, Owner, Related ADRs, Context, Architecture Decisions Made (locked), Architecture Decisions Pending (current focus), Work Done (cumulative, with commit references), Next Steps (ordered, specific), Blockers, NOT Doing (deliberately rejected options), Session Log (append-only).

Discipline rules: update at every session end; read at every session start; locked decisions never reopen; Session Log is append-only; honest status reporting (status doesn’t lie about progress).

See [docs/wip/README.md](#) for the full convention.

2. Data Architecture — Where Data Lives

Data Store	Location	What It Stores	Access
Paperclip DB	Railway PostgreSQL	Agent configs, run history, activity logs, budget spend, issues, approvals, accounting_connections	Scott (admin), all agents via server API
DB Backups	Railway container filesystem	Hourly snapshots — 7 daily, 4 weekly, 1 monthly retained	Scott (admin) only
<code>accounting_connections</code>	Railway PostgreSQL — per-contact rows	Xero/QBO OAuth tokens keyed by (company_id, platform, contact_id)	CFO agents via accounting service
Railway env vars	Railway environment (never committed)	All API keys: GHL, QBO, Xero, encryption keys, webhook secrets, dashboard secret	Railway admin only
GoHighLevel CRM	GHL cloud (SaaS)	Client contacts, pipeline stages, opportunities, forms, communications, tags, custom fields	CRO agents via GHL API, Scott
QuickBooks Online	Intuit cloud — per client instance	Chart of Accounts, transactions, invoices, bills, payroll, reconciliations	CFO agents via QBO OAuth API
Xero	Xero cloud — per client instance	Chart of Accounts, bank feeds, transactions, reports — per-contact OAuth	CFO agents via Xero OAuth API
GitHub Repo	<code>github.com/LedgerixPro/ledgerix-pro-core</code>	All source code — NO secrets, NO .env, NO client data	Scott only (private repo)
Agent AGENTS.md	Git repo — <code>agents/{name}/AGENTS.md</code>	All 18+ agent instructions, SOPs, API call patterns, escalation chains	Scott (author), Paperclip (runtime)

3. Multi-Tenant Architecture (H4-14)

As of H4, Ledgerix Pro supports multiple clients with separate accounting connections in a single Paperclip company. Each client's Xero or QBO connection is keyed by their GHL contact ID, enabling true multi-tenant bookkeeping.

3.1 `accounting_connections` Schema

Column	Type	Purpose
<code>id</code>	UUID	Primary key
<code>company_id</code>	UUID	Always <code>f60117de...</code> — the single Paperclip company
<code>platform</code>	text	<code>'xero'</code> or <code>'quickbooks'</code>
<code>contact_id</code>	text (nullable)	GHM contact ID — NULL for legacy global connection
<code>realm_id</code>	text	Xero tenant ID or QBO company ID
<code>access_token</code>	text (encrypted)	OAuth access token
<code>refresh_token</code>	text (encrypted)	OAuth refresh token
<code>access_token_expires_at</code>	timestamp	Token expiry

Unique constraint: `(company_id, platform, contact_id)` with `NULLS NOT DISTINCT` — one global NULL row per platform and one per-contact row per platform.

3.2 OAuth Connection Flow

Step	Detail
Connect URL	<code>https://api.ledgerixpro.com/api/oauth/xero/connect?contactId={ghlContactId}</code>
State management	<code>pendingStates</code> Map carries <code>contactId</code> across OAuth round-trip (10-min TTL)
Callback	Stores tokens with <code>contact_id</code> in <code>accounting_connections</code>
Agent access	<code>getNewTransactions(db, COMPANY_ID, contact.id, sinceDate)</code> — all agents use this pattern
Portal URL	<code>https://api.ledgerixpro.com/portal/{slug}</code> → 302 redirect to <code>/portal/{contactId}</code>

4. Scheduled Routine Architecture

17 cron routines running on Railway, all in America/Phoenix timezone.

Time (Phoenix)	Cadence	Agent & Routine
6:00 AM	Daily	Sentinel — daily transaction sync for all active clients
6:30 AM	Daily	AP Specialist — daily bill check and overdue flag
7:00 AM	Daily	Tax Liaison — daily deadline check
7:30 AM	Mon-Fri	Quality Control — daily spot-check (20% transaction sample)
7:30 AM	Monday	Reporter — weekly pulse digest
7:00 AM	1st of month	Reporter — monthly deep dive
8:00 AM	Monday	Senior Bookkeeper — weekly digest email to Scott
8:00 AM	1st of month	Billing & Invoicing — monthly client invoicing
8:30 AM	Monday	AP Specialist — weekly AP summary
9:00 AM	Monday	Tax Liaison — weekly deadline summary
9:00 AM	1st of month	Reactivation — monthly churn outreach scan
9:00 AM	Tuesday	Referral & Reviews — weekly 30-day and 90-day candidate scan
9:00 AM	Friday	Quality Control — weekly full review and trend analysis
10:15 AM	Monday	Audit & Compliance — weekly compliance scan
11:00 AM	1st of month	Audit & Compliance — monthly deep scan
11:30 AM	1st of month	Payroll — monthly payroll compliance review
7:00 AM	Bi-weekly Wed	Payroll — bi-weekly payroll run scan

Scale trigger: At 5 paying clients, enable Client Health Monitor weekly heartbeat — Monday 8am Arizona (cron: 0 15 * * 1 UTC). Register via Paperclip routine API at `/companies/f60117de.../routines`.

5. Phase-by-Phase Process Architecture

Phase 1: Lead Generation & Qualification

Sales Division — CRO Oversight

AGENT: SDR — Sales Development Representative (Laura)

Aspect	Detail
Process	Qualifies inbound leads via ICP scoring (1-10 signal confidence). Sends personalized SMS (referral opener) and email sequences by business type. Creates GHL opportunities. Fires <code>contact.sdr_ready</code> webhook on qualification.
Inputs	GHL <code>contact.created</code> or <code>contact.sdr_ready</code> webhook events, ICP qualification criteria, GHL contact custom fields, business type classification
Outputs	ICP score written to GHL custom fields, tags added (<code>icp-qualified</code> , <code>sdr-ready</code>), GHL opportunity created in Sales pipeline, outreach SMS + email sent

AGENT: Client Success Manager

Aspect	Detail
Process	Maintains health scores for all active clients. Delivers plain-English monthly summaries. Monitors churn signals. Runs 30/90/180-day feedback loops.
Inputs	Health score inputs (payment timeliness, response rates, GHL engagement), Reporter financial package, escalations from AR Specialist
Outputs	Client health scores updated in GHL, monthly touchpoint communications, NPS scores routed to CRO and Referral & Reviews, churn risk alerts

AGENT: Referral & Reviews — Tuesday 9:00 AM weekly

Aspect	Detail
Process	Weekly scan identifies 30-day and 90-day candidates. Sends Google review request at 30 days — checks for open issues first, will not ask during unresolved bookkeeping problems. Sends referral ask at 90 days. Uses tags to prevent duplicate outreach.
Inputs	GHL contacts tagged <code>client-active</code> , days-as-client calculation, open issue check, <code>GOOGLE_PLACE_ID</code> env var for review link
Outputs	30-day check-in email with Google review link, 90-day referral email, GHL tags updated (<code>review-requested</code> , <code>referral-requested</code>)

AGENT: Reactivation — 1st of month 9:00 AM

Aspect	Detail
Process	Monitors GHL for churned/paused clients. Classifies churn reason. Executes targeted reactivation sequences with defined hard stops.
Inputs	GHL pipeline stage changes (Paused/Cancelled/Inactive), churn reason, reactivation offers, Do Not Contact list
Outputs	Reactivation sequence messages via GHL, positive response handoffs to Client Success Manager, monthly Churn Intelligence Report to CRO

Phase 2: Client Onboarding

Operations Division — COO Oversight

AGENT: Onboarding — maxTurnsPerRun: 40

Aspect	Detail
Process	Fires on GHL <code>contact.created</code> webhook. Checks for <code>client-active</code> or <code>beta-client</code> tags — if present, escalates instead of re-qualifying. Runs ICP scoring. Sets GHL custom fields (<code>icp_status</code> , <code>signal_confidence_score</code> , <code>client_type</code> , <code>ledgerix_workspace_id</code>). Fires <code>contact.sdr_ready</code> event.
Inputs	GHL <code>contact.created</code> webhook with <code>contactId</code> , GHL contact custom fields, existing tags (<code>client-active</code> , <code>beta-client</code> checks)
Outputs	ICP score written to GHL, tags added, GHL opportunity created, <code>contact.sdr_ready</code> event fired to trigger SDR, escalation issue if existing client detected

AGENT: Knowledge Base Manager

Aspect	Detail
Process	Generates Client Knowledge Package within 24 hours of new client activation. Selects COA template based on client type. Version-controls all documents.
Inputs	Client intake data, COA template library (Trades-HVAC, Trades-Plumbing, Agency, Small Business, Manufacturing), GHL contact ID
Outputs	Client Knowledge Package (client profile, COA template, industry rules, custom instructions) delivered to Senior Bookkeeper

Phase 3: Daily Transaction Processing

Accounting Division — CFO Oversight

AGENT: Sentinel — Daily 6:00 AM Arizona

Aspect	Detail
Process	Pulls all GHL contacts tagged <code>client-active</code> . For each contact, calls <code>getNewTransactions(db, COMPANY_ID, contact.id, yesterday)</code> to fetch Xero/QBO transactions via per-contact OAuth connection. Screens every transaction against 90-day baseline. Flags anomalies, duplicates, and high-risk vendors.
Inputs	GHL contacts tagged <code>client-active</code> , <code>accounting_connections</code> table (filtered by <code>contact_id</code>), Xero/QBO bank feed transactions from yesterday
Outputs	Transaction issues assigned to Ledger Specialist, fraud flags to CFO and COO, runMetrics with <code>transactionsProcessed</code> / <code>autoCategorized</code> / <code>flaggedForReview</code>

AGENT: Ledger Specialist

Aspect	Detail
Process	Assigns COA categories based on vendor history, amount, and industry rules. Standardizes vendor names. Adds job-costing metadata for Trades clients. 80% confidence threshold — below goes to Suspense.
Inputs	Cleaned transactions from Sentinel, Client COA, 12-month vendor history, Client Knowledge Package
Outputs	Categorized transactions posted to QBO/Xero, Suspense-flagged items with notes, processing summary to Senior Bookkeeper

AGENT: Senior Bookkeeper — Monday 8:00 AM weekly digest

Aspect	Detail
Process	Reviews Ledger Specialist output for GAAP compliance. Detects anomalies >20% variance from 3-month rolling average. 95% confidence threshold. Sends weekly digest email to scott@ledgerixpro.com every Monday 8am.
Inputs	Categorized transactions from Ledger Specialist, prior period financial data, client Chart of Accounts
Outputs	Verified transactions, GAAP-compliant entries, HITL flags with Reason for Doubt, weekly digest email

Phase 4: Accounts Payable & Receivable

Accounting Division — CFO Oversight

AGENT: AP Specialist — Daily 6:30 AM + Monday 8:30 AM

Aspect	Detail
Process	Records all incoming vendor bills in QBO/Xero. Routes bills above approval threshold for human sign-off. Schedules payments 2 business days before due date. Flags duplicates and new unapproved vendors.
Inputs	Vendor bills from QBO/Xero, client approval thresholds, approved vendor list from Knowledge Base Manager
Outputs	Bills entered in QBO/Xero, approved payments scheduled, AP Aging Report at month-end, weekly summary email

AGENT: AR Specialist (with Payment Date Intelligence)

Aspect	Detail
Process	Generates invoices, applies payments, executes collections sequence. Payment Date Intelligence: captures confirmed dates from client email replies ('I'll pay by Friday'), uses 12-invoice history to predict payment dates, recalibrates after payment received. Manual override via <code>PAYMENT_DATE: YYYY-MM-DD</code> in issue comments. Escalation: Senior Bookkeeper → CFO → Scott.
Inputs	Work orders, GHL triggers, incoming payment notifications, client email replies with payment commitments
Outputs	Invoices sent via QBO/Xero, payments matched, AR Aging Report, collections sequence via GHL, payment date predictions in QBO/Xero

AGENT: Payroll — Bi-weekly Wednesday 7:00 AM + Monthly 1st 11:30 AM

Aspect	Detail
Process	Monitors payroll expenses posted to QBO/Xero — does not run payroll directly (clients use Gusto/ADP/Paychex). Checks 941 deposit timing, Arizona withholding, W-2 deadlines, FUTA, 1099 vs W-2 misclassification risk. Escalation: Senior Bookkeeper → CFO → Scott.
Inputs	QBO/Xero payroll expense transactions, payroll tax deposit tracking, client payroll provider
Outputs	Bi-weekly scan results, monthly payroll report email, compliance flags to Senior Bookkeeper, W-2/FUTA deadline alerts in December/January

Phase 5: Month-End Reconciliation & Close

Accounting Division — CFO Oversight

AGENT: Reconciliation Agent

Aspect	Detail
Process	Verifies categorized transactions match official bank statements to the penny. Applies the Penny Rule (\$0.00 variance required). Flags stale items, duplicates, and unexplained variances.
Inputs	Official bank and credit card statements from QBO/Xero, categorized transactions from Ledger Specialist, prior period closing reconciled balance
Outputs	Reconciliation Report (\$0.00 variance confirmed or blocked), cleared transaction confirmations, escalations to Sentinel for variances >\$500

AGENT: Quality Control — Daily 7:30 AM Mon-Fri + Friday 9:00 AM weekly

Aspect	Detail
Process	Daily spot-check: 20% transaction sample across four dimensions (categorization accuracy, duplicate detection, anomaly detection, KB rule coverage). Friday weekly full review with trend analysis and recommendations report emailed to scott@ledgerixpro.com. Escalation thresholds: error rate >5% → Senior Bookkeeper (urgent), KB coverage <50% → KB Manager.
Inputs	Ledger Specialist completed issues, prior QC runMetrics, KB Manager activity logs
Outputs	Daily QC spot-check results, Friday QC weekly report email, QC Alert issues, runMetrics with <code>errorRate / autoCategorizationRate / hit1EscalationRate</code>

AGENT: Reporter — Monday 7:30 AM + 1st of month 7:00 AM

Aspect	Detail
Process	Weekly pulse Monday morning + monthly deep dive on the 1st. Generates P&L, Balance Sheet, Cash Flow Summary, AR Aging Report, and plain-English CFO-level insight per client.
Inputs	QC-approved reconciled data from QBO/Xero, Senior Bookkeeper sign-off, client business type
Outputs	Monthly financial package to CFO, plain-English summary for client portal, weekly pulse email to scott@ledgerixpro.com

Phase 6: Tax & Compliance

Accounting Division — CFO Oversight

AGENT: Tax Liaison — Daily 7:00 AM + Monday 9:00 AM

Aspect	Detail
Process	Maintains federal and Arizona state tax deadline calendar. Alerts 30 days before every deadline. IRS: 1099-NEC, estimated quarterly payments, 941 deposits. Arizona: TPT, A1-QRT, estimated tax (4.5%). Prepares CPA data packages.
Inputs	Reconciled financial data from QBO/Xero, client tax jurisdiction (federal + Arizona), payroll tax alerts from Payroll Agent
Outputs	Daily/weekly deadline alerts, 30-day and 7-day warnings, CPA data packages, year-end close checklist

AGENT: Audit & Compliance — Monday 10:15 AM + 1st of month 11:00 AM

Aspect	Detail
Process	Weekly scan + monthly deep scan. Covers IRS (1099, estimated taxes, expense documentation, 50% meal limit), Arizona (TPT, ROC contractor licenses, business licenses), and industry-specific compliance (trades ROC expiration, retail TPT, agency E&O insurance). Escalation: Senior Bookkeeper → CFO → Scott. Never emails Scott for routine flags.
Inputs	GHL contacts tagged <code>client-active</code> with <code>client_type</code> , QBO/Xero vendor payment history, GHL contact notes (ROC license, insurance docs)
Outputs	Weekly compliance scan results, monthly compliance report email to scott@ledgerixpro.com , Paperclip issues to Senior Bookkeeper for each compliance finding

Phase 7: Internal Revenue Operations

Internal Operations — CEO/CFO Oversight

AGENT: Billing & Invoicing — 1st of month 8:00 AM

Aspect	Detail
Process	Manages Ledgerix Pro's own revenue. Sends monthly service invoices to all active clients. Service tiers: The Foundation (\$199/\$299), The Growth Engine (\$399/\$599), The Scale-Up (\$999/\$1,299). First 10 clients receive Charter pricing.
Inputs	Active client roster from GHL (<code>client-active</code> tag), service tier custom field, contracted rates
Outputs	Monthly invoices, MRR report to CFO and CRO, revenue leakage risk alerts, Day 30 non-payment escalations to CFO

6. Phase 4 — Accounting API & Phase 4c Safety Architecture

Phase 4 builds the HTTP API endpoint layer atop the per-tenant accounting service infrastructure shipped in H2-H4. Pattern B Full (per ADR-001, May 17, 2026) is the primary agent interface — agents call `/api/accounting/v1/*` endpoints rather than invoking service functions directly through scripts.

6.1 Endpoint Roster

Endpoint	Status	Notes
<code>GET /transactions</code>	Production-ready	List with pagination, filtering
<code>GET /transactions/:txnId</code>	Partial (Q3 pending)	Needed by write endpoints for <code>previousAccountRef</code> capture
<code>GET /accounts</code>	Production-ready	Chart of Accounts retrieval
<code>GET /invoices</code>	Production-ready	List + filtering
<code>GET /reports/p-and-l</code>	Production-ready	P&L with Balance Sheet + Trial Balance extensions
<code>POST /transactions/:txnId/category</code>	DEFERRED (Phase 4c.5)	Atop safety layer, blocked on Q3 resolution
<code>POST /payments</code>	DEFERRED (Phase 4c.5)	Atop safety layer (thresholds)
<code>POST /invoices</code>	DEFERRED (Phase 4c.5)	Atop safety layer (pricing + dedupe), blocked on Q1 + Q2

5 of 8 endpoints production-ready. Write endpoints intentionally deferred behind Phase 4c safety architecture per the Trust Tenet.

6.2 Phase 4b — Write Endpoint Foundation

Three foundational pieces shipped May 23-24 before any write endpoint:

- **Idempotency** via `idempotency_keys` table + `withIdempotency` wrapper. Replays return cached response forever regardless of approval status.
- **Audit logging** via extended `activity_log` table with `status` column (success | failure). Every operation tied to actor identity.
- **Two-phase failure handling** per ADR-002 D2: upstream call inside work callback; audit log success AFTER upstream success; audit log failure BEFORE returning 502.

6.3 Phase 4c — Safety Architecture

Every write endpoint sits atop a safety layer that intercepts financial operations and either auto-proceeds (high-confidence cases) or creates a human-in-the-loop approval request (lower-confidence or threshold-breaching cases). This is the operationalization of the Trust Tenet for write operations.

Endpoint flow:

1. Validate input
2. Auth check
3. Safety layer checks: pricing match, threshold check, dedupe check, previous-state capture
4. If ANY check returns “approval required” → create approval row, return HTTP 202 Accepted
5. If ALL checks pass → upstream write, audit log success, return 200

6.3.1 Pricing Source of Truth (Phase 4c.1, shipped 2026-05-24)

Schemas:

- `service_tier_pricing` — canonical (tier, isCharter) → monthly amount in cents. Effective-dated.
- `client_pricing_overrides` — per-client deviations from canonical. Effective-dated.

Service: `getExpectedPriceCents(db, tier, isCharter, contactId?)` returns the expected monthly amount, applying client-specific override if present.

Migration: `0065_whole_post.sql`

6.3.2 Threshold Framework (Phase 4c.2, shipped 2026-05-24)

Schema: `write_thresholds` — (endpoint, field, comparator, threshold_value, action, reason) tuples. Each can be global (NULL `ghl_contact_id`) or per-client. Effective-dated. Hierarchical: per-client overrides global; per-client can either tighten or loosen vs global default.

Services:

- `getApplicableThresholds(db, endpoint, field, contactId)` — returns matching thresholds (global + per-

client)

- `getMostSpecificThreshold(db, endpoint, field, contactId)` — returns the threshold that applies
- `isThresholdExceeded(value, threshold)` — comparator evaluation (gt, gte, lt, lte, eq)

Bootstrap data (DEFERRED to Phase 4c.5 admin endpoint runbook):

- Global: `accounting.payments` field `amount` gt 1,000,000 cents = \$10K — per Section 7.3 HITL Gates
- Global: `accounting.invoices` field `lineItems.sum` gt 100,000 cents = \$1K — conservative anomaly default

Migration: `0066_familiar_nighthawk.sql`

6.3.3 Customer Dedupe with HITL (Phase 4c.3, shipped 2026-05-24)

Refactored QBO `findOrCreateCustomer` returns `{customerId, action, matchDetails?}` instead of silent string. Five action types:

Action	Meaning	Auto-proceed
<code>found_by_email</code>	Email match; names similar or empty	Yes
<code>found_by_name_exact</code>	Name match; no email conflict	Yes
<code>created_new</code>	No match found; created	Yes
<code>ambiguous_email_match_different_name</code>	Email matches but names differ significantly	No — approval required
<code>ambiguous_name_only</code>	Name matches but emails differ (both present)	No — approval required

New utility module: `services/accounting/string-similarity.ts` (`normalizeName` , `levenshteinDistance` , `namesAreSimilar`). Default similarity threshold 3 Levenshtein edits, configurable via `LEDGERIX_NAME_SIMILARITY_THRESHOLD` env var.

When action is HITL-required, caller MUST NOT proceed with the upstream write. Caller creates an approval row with `accounting.invoice.dedupe_ambiguous` type and waits for human resolution.

6.3.4 Write-Approval Dispatcher (Phase 4c.4, shipped 2026-05-24 in stub mode)

When a write endpoint detects a safety condition requiring approval, it creates an approval row instead of executing the write. When the approval is later approved by a human, the dispatcher executes the deferred write using the payload data.

Module: `services/accounting/write-approvals.ts`

Four approval types (dot-namespaced per ADR-003 Q1):

- `accounting.payment.threshold_exceeded`
- `accounting.invoice.dedupe_ambiguous`
- `accounting.invoice.pricing_mismatch`
- `accounting.transaction.category_with_unknown_previous`

Each type has a TypeScript payload interface capturing all data needed to execute the deferred write.

Dispatcher integration: `approvalService.approve()` extended to call `executeApprovedAccountingWrite(db, approval)` when approval type matches `accounting.*`. Dispatcher failures do NOT roll back the approval — the approval IS approved; the downstream write either succeeded, will be retried, or failed in ways needing human follow-up.

Stub mode: Phase 4c.4 ships the routing infrastructure; each case logs the approved-write event but does NOT execute the upstream QBO/Xero write. Phase 4c.5 will replace each stub case with the actual upstream call.

6.3.5 Write Endpoint Re-Implementation (Phase 4c.5, IN PROGRESS)

Re-ship `POST /transactions/:txnId/category` , `POST /payments` , `POST /invoices` atop the complete safety layer. Wire Phase 4c.4 dispatcher stubs to real upstream writes.

Part 1 shipped (commit `e618231b` , 2026-05-24):

- Migration `0067_last_gateway.sql` — `activity_log.company_id` nullable for system-scoped admin

operations

- `LogActivityInput` type accepts `companyId: string | null`
- Live-events and plugin-events suppressed when `companyId` is null
- Generic helper `services/admin/compare-and-seed.ts` for version-aware idempotent seeding
- WIP doc shipped at `docs/wip/phase-4c-5-write-endpoints-and-admin-api.md`

Architecture Decisions locked in Phase 4c.5 (in WIP doc):

Decision	Locked	Summary
1	2026-05-24	Admin HTTP endpoints (not one-time scripts) for safety-layer data management. Required for 7-year audit retention.
2 (revised)	2026-05-24	Admin endpoints use existing <code>assertInstanceAdmin</code> from <code>authz.ts</code> (natively supports session, board_key, local_implicit — all identity-tracked).
3	2026-05-24	Version-aware idempotency (Option D-modified). Identical → skip; different → supersede with effective-dating; missing → insert.
B	2026-05-24	<code>activity_log.company_id</code> nullable for system-scoped admin operations.
Option 1	2026-05-24	Live-events and plugin-events suppressed when <code>companyId</code> is null.

Architecture questions pending (each blocks specific endpoints):

- **Q1: Charter status storage** — blocks Invoice endpoint. `getExpectedPriceCents` requires `isCharter` parameter; no defined storage exists yet.
- **Q2: Setup fee handling** — blocks Invoice endpoint. Setup fees (\$249/\$349/\$1,200 per Section 7) not modeled by current pricing schema.
- **Q3: get-transaction-by-id infrastructure scope** — blocks Transaction Category endpoint. Returning `previousAccountRef` requires fetching transaction by ID; no such function exists per QBO/Xero transaction type.

Explicitly rejected (NOT Doing):

- One-time scripts for seeding (7-year audit retention requires durable `activity_log`)
- Every-category-update-needs-approval (doesn't scale to 50+ clients)
- `previousAccountRef: null` placeholder (Trust Tenet)

6.4 Architectural Decision Records

Phase 4 architectural decisions are captured in ADR documents under `docs/adr/`:

- **ADR-001** — Pattern B Full API endpoints as primary agent interface
- **ADR-002** — Phase 4b write endpoint design (idempotency, audit, two-phase failure)
- **ADR-003** — Phase 4c safety architecture (10 decisions + 3 amendments)
- **ADR-004** — Pending; will capture Phase 4c.5 locked decisions when Phase 4c.5 ships

7. Data Security & Guardrails

7.1 The Four Core Security Guardrails

Guardrail	How It Works	Why It Matters
1. Data Isolation	Each client keyed by GHL contact ID in <code>accounting_connections</code> . All agent queries filter by <code>contact_id</code> . No agent can access another client's data.	Company A's books never touch Company B's data. A breach is a Level 1 emergency.
2. Access Limits	QBO and Xero OAuth scopes: read-only for reports, read/write for transactions. Money movement is never requested in OAuth scopes.	Even a compromised agent cannot move money. Worst case: a miscategorized transaction.
3. Token Budgeting	Agent budget caps enforced in Paperclip. Current beta caps ~\$215/mo for 2 clients. Scale targets: ~\$1,225 at 25, ~\$3,110 at 75, ~\$5,480 at 150, ~\$9,700 at 300.	Prevents AI cost runaway. Revisit at each 25-client milestone.
4. Human Gate	No transaction marked Verified below 95% Senior Bookkeeper confidence. All HITL flags held for human review in dashboard queue.	AI does the work, humans own the outcome. Core trust mechanism for clients.

7.2 Escalation Chain

Standard escalation chain across all Ledgerix Pro agents:

Senior Bookkeeper → CFO → Scott

No agent contacts Scott directly for routine operational issues. Scott is notified only when the escalation chain is exhausted or for Level 3 events (confirmed fraud, data breach, client cancellation).

7.3 Human-in-the-Loop Gates

Gate	Trigger / Condition
95% confidence threshold	Senior Bookkeeper flags any transaction below 95% confidence with Reason for Doubt
\$0.00 reconciliation close	Senior Bookkeeper and CFO must sign off before period is declared closed
New Chart of Accounts category	No new COA category created without human administrator approval
Payroll runs >\$10,000	CFO must sign off before processing
Phase 4c safety layer	Every write endpoint operation passes through pricing / threshold / dedupe / previous-state checks. Any “approval required” outcome creates an approval row (per ADR-003 Q1 type taxonomy) and returns HTTP 202 — actual upstream write deferred until human approval.
Client cancellation	CEO and human admin must lead — no autonomous retention
Level 3 financial risk	Confirmed fraud or material misstatement — immediate CEO notification, workspace paused
Data isolation breach	Any cross-client data event — Level 1 emergency, CEO + human admin notified immediately
Budget cap exceptions	CEO must authorize any exception to per-client budget caps
Reactivation offers	CRO must authorize all win-back offers with defined expiration dates

7.4 API Key & Secret Management

Secret	Storage	Protection Method
<code>ANTHROPIC_API_KEY</code>	Railway env var (no quotes)	Never committed. Critical: no quotes in Railway env vars — quoted values cause auth failures.
<code>GHL_API_KEY</code>	Railway env var	Never committed. Rotated every 90 days.
<code>GHL_WEBHOOK_SECRET</code>	Railway env var	Shared secret — <code>X-Ledgerix-Secret</code> header on every inbound GHL webhook
<code>ENCRYPTION_KEY</code>	Railway env var	32-byte hex key for encrypting OAuth refresh tokens at rest
<code>DASHBOARD_SECRET</code>	Railway env var	SHA-256 hash — protects <code>/dashboard</code> route
<code>QBO_CLIENT_ID/SECRET</code>	Railway env var	OAuth 2.0 for QuickBooks Online
<code>XERO_CLIENT_ID/SECRET</code>	Railway env var	OAuth 2.0 for Xero — LIVE
<code>DATABASE_URL</code>	Railway env var (internal)	Railway internal Postgres connection string
<code>GOOGLE_PLACE_ID</code>	Railway env var (pending)	Added when Google Business profile is claimed — used by Referral & Reviews agent

7.5 Safety Layer (Phase 4c)

Position in stack: between HTTP route handler and QBO/Xero service call. Every write endpoint must traverse this layer. See Section 6.3 for full architecture.

Mechanism: each write endpoint, before executing the upstream write, queries the safety layer for pricing check, threshold check, dedupe check, and previous-state capture. If ANY check returns “approval required,” the endpoint creates an approval row with the full request payload and returns HTTP 202 Accepted. The actual upstream write is deferred until a human approves. The Phase 4c.4 dispatcher executes the deferred write when approval is granted.

Why this matters: AI agents can make write requests at machine speed; the safety layer ensures human review on anything that materially affects client books — without making EVERY write require approval (which would not scale to 50+ clients).

8. Service Tiers & Pricing

Updated May 17, 2026 (v3.2). Growth Engine Standard moved from \$499 to \$599 and Scale-Up repriced to \$999/\$1,299 to match market positioning for specialist trades and agency bookkeeping. First 10 clients receive Charter pricing; the Charter benefit follows the client across tier upgrades and downgrades for as long as service is continuous (see Section 8.1 Charter Pricing Window). All pricing is monthly recurring.

Tier	Target Client	Charter	Standard	Key Features
The Foundation	Freelancers & Micro-business	\$199/mo	\$299/mo	Full bookkeeping, monthly P&L, tax deadline alerts
The Growth Engine	Local Trades & Service businesses	\$399/mo	\$599/mo	Foundation + job costing, ROC license tracking, TPT compliance
The Scale-Up	Agencies & Law Firms	\$999/mo	\$1,299/mo	Growth Engine + trust account compliance, E&O tracking, multi-entity

Setup & Migration Fee

All clients (including Charter) pay a one-time setup fee at onboarding. The fee covers chart of accounts review, vendor categorization rules, platform connection, workflow training, and (for prospects not currently on QBO or Xero) data migration. No waivers.

Tier	Setup Fee
The Foundation	\$249
The Growth Engine	\$349
The Scale-Up	\$1,200

The setup fee applies once at onboarding. It is non-refundable except per the 30-day satisfaction guarantee (see Strategic Plan).

8.1 Tier Qualifiers

Industry framing drives the marketing pitch; the qualifiers below drive the price quoted at the discovery call. Most qualifiers must match to land a client in a given tier. If a client splits across tiers (for example, Growth Engine on transaction volume but Scale-Up on entity count), price to the higher tier and call out the upgrade rationale in the proposal.

Qualifier	Foundation	Growth Engine	Scale-Up
Monthly transaction volume	Up to 75	75-300	300+
Bank/credit accounts	1-2	2-5	5+
Employees on payroll	0-2	3-15	10-25
Accounting integrations	1 platform	2-3 platforms (FSM, etc.)	3+ platforms
Annual revenue	Under \$500K	\$500K-\$3M	\$2M+
Job costing required	No	Yes	Yes
Trust/multi-entity	No	No	Yes
Industry flags	None	ROC, TPT, seasonal	IOLTA, E&O, accrual

Charter Pricing Window

Charter pricing applies to the **first 10 paying clients only**. After client #10, the Charter column closes and new clients onboard at Standard pricing.

Charter benefit follows the client across tiers. As long as service is continuous, Charter clients keep Charter pricing for the duration of their relationship with Ledgerix Pro — even as their business grows from one tier to the next. A Foundation Charter client whose qualifiers move them to Growth Engine pays Growth Engine Charter (\$399/mo), not Standard (\$599/mo). Same applies for Scale-Up: Charter clients who grow into Scale-Up pay Scale-Up Charter (\$999/mo), not Standard (\$1,299/mo). The benefit also applies on tier downgrades: a Scale-Up Charter client whose qualifiers shrink to Growth Engine pays Growth Engine Charter, not Standard.

Continuity rule. If a Charter client cancels service and later returns, they re-enter at Standard pricing for whichever tier they qualify for. The Charter benefit is granted to the first 10 paying clients as a thank-you for choosing Ledgerix Pro early; it is not transferable, sellable, or recoverable after cancellation.

8.2 Scale Milestones & Triggers

Milestone	Action Required
5 paying clients	Enable Client Health Monitor weekly heartbeat: cron <code>0 15 * * 1</code> (Monday 8am Arizona / 15:00 UTC)
10 clients	Charter pricing window closes — new clients at Standard pricing
Each 25-client milestone	Revisit agent budget caps. Targets: ~\$1,225 at 25, ~\$3,110 at 75, ~\$5,480 at 150, ~\$9,700 at 300
~50 leads	Enable Sales Outreach agent (H4-9)
~50 clients	Implement Scale Pattern for accounting webhooks (QBO webhook renewal cron, queue-based burst processing, dead letter handling)
~50 clients	Migrate KB rules to database table for faster agent lookup (H4-12/13)

9. Client Portal & Dashboard

9.1 Client Portal

Property	Detail
URL format (contact ID)	<code>https://api.ledgerixpro.com/portal/{ghlContactId}</code>
URL format (slug)	<code>https://api.ledgerixpro.com/portal/{company-slug}</code> → 302 redirect to contact ID URL
Slug generation	Company name slugified: spaces → hyphens, lowercase, special chars stripped. 'Acme Industries' → 'acme-industries'
Authentication	None for beta — <code>contactId</code> is effectively the token. Add proper auth before paying clients request it.
Current beta clients	None — pending Pattern B Full Phase 5

9.2 Internal Dashboard

Property	Detail
URL	<code>https://api.ledgerixpro.com/dashboard</code>
Authentication	Dashboard secret (SHA-256 hash) entered on load
Content	Agent health status, HITL queue, active clients list, last run times, issue counts
Summary API	<code>https://api.ledgerixpro.com/dashboard/summary</code> (JSON)
Agent health statuses	idle / running / degraded (timeout count >5 in recent runs)

10. Live Integration Status (May 2026)

Integration Component	Status	Detail
Railway deployment	LIVE	<code>api.ledgerixpro.com</code> — auto-deploys on git push to master
GHL webhook pipeline	LIVE	<code>contact.created</code> , <code>contact.updated</code> , <code>contact.sdr_ready</code> all routing correctly
Xero OAuth (multi-tenant)	LIVE	Per-contact connections via <code>/oauth/xero/connect?</code> <code>contactId=...</code>
QBO OAuth (multi-tenant)	LIVE	Built and deployed — pending first QBO client
Xero webhook	LIVE	Registered at developer.xero.com — Invoices, Contacts, Credit Notes, Billing Subscriptions. Status: OK.
All 17 cron routines	LIVE	Firing on schedule — Sentinel, AP Specialist, Tax Liaison confirmed running daily
Client portal (slug-based)	LIVE	Slug redirect working: <code>/portal/{company-slug}</code> → 302 → <code>/portal/{contactId}</code>
Scanner path hardening	LIVE	<code>/.env</code> , <code>/.git/*</code> , <code>/.aws/*</code> , URL-encoded variants all returning 404
DB hourly backups	LIVE	Confirmed running — ~296KB snapshots every hour, 7-day daily retention
A2P SMS messaging	PENDING	GHL Trust Center approval pending — Laura's SMS outreach queued
Google Business Profile	PENDING	<code>GOOGLE_PLACE_ID</code> to be added to Railway after profile is claimed
Client portal auth	DEFERRED	Add proper token auth when first paying client requests portal access

11. Railway Deployment Reference

11.1 Key Infrastructure Facts

- Railway auto-deploys on every push to master branch of `github.com/LedgerixPro/ledgerix-pro-core`
- Dockerfile: `node:22-bookworm-slim` , non-root paperclip user (UID 1001), Claude CLI 2.1.138
- **CRITICAL:** Railway env vars must NOT have quotes. `KEY=value` , never `KEY="value"` . Quoted values caused `ANTHROPIC_API_KEY` auth failures.
- Migrations apply automatically on container boot via Paperclip's migration runner
- If Railway does not auto-redeploy after push: use an empty commit to trigger (`git commit --allow-empty`)

11.2 GHL Custom Fields

Field Name	Internal ID	Purpose
<code>service_tier</code>	<code>Dh5rwdlahz6a37BAQDI</code> s	Client service tier (The Foundation / The Growth Engine / The Scale-Up)
<code>client_type</code>	<code>Cf539co3LHJrm6wLAJQJ</code>	Business type (manufacturing, trades, agency, law_firm, retail, etc.)
<code>ledgerix_workspace_id</code>	<code>vmAT40jG10QboXA2Jqjs</code>	Paperclip company ID (vestigial post-H4-14 — retained for backward compat)
<code>nurture_month</code>	<code>sMQegZrU2giDsyaNKnjt</code>	Nurture sequence month tracking

11.3 GHL Pipeline IDs

Pipeline	ID
Sales Pipeline	<code>dtgrQV0u9DB5EmxJGY9K</code>
Clients Pipeline	<code>EOq8U8BCqRMX9kM5g2qS</code>
Churn Pipeline	<code>A4SSmXmDnwPKGfxKcvut</code>

11.4 Beta Client Reference

Beta clients: None active. Enyrgy Inc will be onboarded through the production pipeline (`api.ledgerixpro.com/free-audit` → Tier-Fit Audit → standard onboarding flow) once Pattern B Full Phase 5 ships.

11.5 Key File Locations

File / Directory	Purpose
<code>agents/{name}/AGENTS.md</code>	Agent instructions — 18 active agents, each with full SOP
<code>server/src/routes/oauth/xero.ts</code>	Xero OAuth routes — per-contact connect and callback
<code>server/src/routes/oauth/quickbooks.ts</code>	QBO OAuth routes — per-contact connect and callback
<code>server/src/routes/webhooks/ghl.ts</code>	GHl webhook receiver with dual-path auth
<code>server/src/routes/ledgerix-dashboard.ts</code>	Dashboard, portal, and slug redirect routes
<code>server/src/services/accounting/index.ts</code>	<code>getNewTransactions</code> — primary accounting data access function
<code>packages/db/src/schema/accounting_connections.ts</code>	Multi-tenant accounting connections schema
<code>RESET.md</code>	Complete recovery runbook — 31 sections including backup recovery (Section 31)
<code>.env.example</code>	Safe placeholder template for all Railway environment variables

11.6 Database Migrations

Migration head: `0067_last_gateway.sql`

Recent migrations relevant to Phase 4:

Migration	Purpose	Phase
0064_aged_iron_man	activity_log.status column + idempotency_keys table	Phase 4b foundation
0065_whole_post	service_tier_pricing + client_pricing_overrides	Phase 4c.1
0066_familiar_nighthawk	write_thresholds	Phase 4c.2
0067_last_gateway	activity_log.company_id nullable	Phase 4c.5 Decision B

Migration workflow: edit schema TypeScript file → `pnpm generate` → drizzle-kit auto-creates SQL + snapshot. Apply automatically on container boot via Paperclip migration runner.

11.7 Test Coverage Baseline

Current targeted test count: **145 tests passing**

Domain	Tests	Phase
Accounting routes	62	Phase 4a
Idempotency	14	Phase 4b
Pricing	10	Phase 4c.1
Thresholds	14	Phase 4c.2
String similarity	25	Phase 4c.3
findOrCreateCustomer	9	Phase 4c.3
Write-approvals dispatcher	11	Phase 4c.4

Targeted vitest runs (`pnpm exec vitest run <file>`) are reliable. Full server suite (`pnpm exec vitest run` — 1240 tests) has known flakiness in `src/__tests__/workspace-runtime.test.ts` due to parallel-run resource contention. Verified 2026-05-24: same tests pass cleanly when run in isolation. Pre-existing infrastructure issue unrelated to current work.

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Scott Hansbury | Founder & Business Strategist | Scottsdale, Arizona | Version 3.4 | May 24, 2026